

ORBITAL COLLISION ASSESSMENT

AEGIS ORBITAL — CLASSIFIED REPORT

Generated: Sat, 29 Aug 2026 09:47:21 GMT

Prediction Window: +49 hours

SATELLITE IDENTIFICATION

Name: **STARLINK-1007**
NORAD ID: **44713**
Type: **SATELLITE**
Data Source: **active_satellites_1771840820649.txt**
Status: **OPERATIONAL**

TWO-LINE ELEMENT (TLE) DATA

1 44713U 19074A 26080.95622857 .00000964 00000+0 30925-3 0 9998
2 44713 53.0540 12.4349 0057662 275.5770 83.8855 15.06000000220001

CONJUNCTION ANALYSIS

3 potential conjunction event(s) detected within 49-hour window

#	DEBRIS / OBJECT	OBJECT ID	TIME OF CLOSEST APPROACH (UTC)	MISS DIST. (km)	REL. SPEED (km/s)	RISK
1	COSMOS 2251 DEB (FRAG 1)	34123	2026-08-29 09:47:21	0.082	14.32	CRITICAL
2	FENGYUN 1C DEB	786	2026-08-29 09:47:21	13.830	2.58	WARNING
3	FENGYUN 1C DEB	753	2026-08-29 09:47:21	42.280	14.85	WARNING

RISK ASSESSMENT & METHODOLOGY

Aegis Orbital determines potential close approaches by propagating the primary and secondary objects using **SGP4** and calculating their relative separation over the prediction window. For each candidate pair: $\mathbf{r}_{rel} = \mathbf{r}_{secondary} - \mathbf{r}_{primary}$, miss distance $d = ||\mathbf{r}_{rel}||$, and relative velocity $\mathbf{v}_{rel} = ||\mathbf{v}_{secondary} - \mathbf{v}_{primary}||$. The system identifies the minimum predicted separation and corresponding Time of Closest Approach (TCA).

Standard TLE data does not contain complete orbital covariance information. Therefore, this report provides a simplified conjunction-risk assessment rather than an operational probability of collision. Professional collision probability assessment requires higher-fidelity state information and uncertainty/covariance data.

ANALYSIS PARAMETERS & HIGHEST PRIORITY EVENT

ANALYSIS PARAMETERS	HIGHEST PRIORITY CONJUNCTION
Propagation Model: SGP4 Prediction Window: +49 hours Screening Threshold: 50.0 km Propagation Step: 60 seconds Coordinate System: TEME / ECEF Data Epoch: 2026-08-29 09:47:21 UTC	Primary: STARLINK-1007 Secondary: COSMOS 2251 DEB (FRAG 1) TCA: 2026-08-29 09:47:21 Miss Distance: 0.082 km Relative Speed: 14.32 km/s (Risk: CRITICAL) Priority is elevated due to the small predicted miss distance combined with high relative velocity.